NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 11

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

What are some of the other names of the smart grids?

a. Electricity with a brain
b. Energybar
c. Electrolyte
d. All of these

Correct Answer: a. Electricity with a brain

Detailed Solution: Smart grids are also known as Electricity with a brain, Energy internet, and Electronet. (Please refer Lecture 51@8:33)

QUESTION 2:

Traditional Grid follows which type of flow of energy?

a. Unidirectional

b. Bidirectional

c. Both a and b

d. None of these

Correct Answer: a. Unidirectional

Detailed Solution: Traditional grid follows unidirectional flow of energy. (Please refer Lecture 51@5:41)

QUESTION 3:

Which of the following is a benefit associated with smart grids?

a. Efficient transmission of electricity
b. Lower electricity rates
c. Improved security
d. All of these

Correct Answer: d. All of these

Detailed Solution: Smart Grids provide efficient transmission of electricity, lower electricity rates and Improved security. (Please refer Lecture 51@11:26)

QUESTION 4:

Which of the following is a way of consumer engagement?

a. Net metering
b. Time-of-Use pricing
c. Financial incentives
d. All of these

Correct Answer: d. All of these

Detailed Solution: Net metering, Time-of-Use pricing and Financial incentives are ways of consumer engagements. (Please refer Lecture 51 @32:31)

QUESTION 5:

Fill in the blank.

In _____, a home can have power from distributed resources.

a. Peaking
b. Islanding
c. Neither a nor b

d. Both a and b

Correct Answer: b. Islanding

Detailed Solution: In islanding, a home can have power from distributed resources.(Please refer Lecture 51 @31:20)

QUESTION 6:

Which of the following aggregates energy consumption of a certain geographical area?

a. Data Aggregator Units
b. Meter Data Management System
c. Buffer

d. None of these

Correct Answer: a. Data Aggregator Units

Detailed Solution: Data Aggregator Units aggregates energy consumption of a certain geographical area (Please refer Lecture 52@15:03)

QUESTION 7:

Which of the following statements are true about Dynamic System Attacks?

Statement I: Replay attacks compromises sensors and monitors their outputs.

Statement II: D-DIA can lead to a system collapse.

Statement III: Covert attack is a closed loop version of replay attacks.

a. Statements I and II
b. Statements I and III
c. Statements II and III
d. Statement I, II and III

Correct Answer: d. Statements I, II and III

Detailed Solution: Replay attacks compromises sensors and monitors their outputs. D-DIA can lead to a system collapse. Covert attack is a closed loop version of replay attacks. (Please refer Lecture 52@20:03)

QUESTION 8:

Fill in the blank.

_____ is a centralized coordinator for smart grid communication.

a. Gateway
b. Meter
c. AMH
d. MDMS

Correct Answer: d. MDMSs

Detailed Solution: MDMS is a centralized coordinator for smart grid communication. (Please refer Lecture 52@15:32)

QUESTION 9:

Fill in the blank.

_____ are available protocols for smart home appliances.

a. C-Bus
b. DECT
c. EnOcean
d. All of these

Correct Answer: d. All of these

Detailed Solution: C-Bus, DECT, EnOcean and Universal Powerline Bus are available protocols for smart home appliances. (Please refer Lecture 52@12:35)

QUESTION 10:

Fill in the blank. _____ is the time required to transmit the data.

a. Scalability

b. Topology

c. Throughput

d. Latency

Correct Answer: d. Latency

Detailed Solution: Latency is the time required to transmit the data. (Please refer Lecture 53@21:20)

QUESTION 11:

IIoT is based on Wrap & Replace approach.

a. True
b. False

Correct Answer: b. False

Detailed Solution: IIoT is based on wrap & re-use approach. (Please refer Lecture 53@9:53)

QUESTION 12:

Which of the following happened in the 3rd Revolution?

a. Mass production
b. IIoT
c. Mechanized production
d. None of these

Correct Answer: d. None of these

Detailed Solution: Internet evolution and automation happened in the 3rd Revolution.(Please refer Lecture 53@10:06)

QUESTION 13:

Which of the following is used to improve the interoperability of different systems?

a. Standardization
b. Optimization
c. Monitoring
d. None of these

Correct Answer: a. Standardization

Detailed Solution: Standardization improves the interoperability of different systems. (Please refer Lecture 54@16:09)

QUESTION 14:

Big data is characterized by 7 Vs.

a. True
b. False

Correct Answer: a. True

Detailed Solution: Big data is characterized by 7 Vs. (Please refer Lecture 55@10:36)

QUESTION 15:

What is Hadoop used in?

a. Distributed processing of large datasets
b. Small clusters of computers
c. Both a and b
d. None of these

Correct Answer: a. Distributed processing of large datasets

Detailed Solution: Hadoop is used for distributed processing of large datasets across Large clusters of computers (Please refer Lecture 55@23:18)

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